

University of Eloued
Academic Year: 2025/2026

Lesson: Linear Regression and Logistic Regression

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Short Summary and Series

Series 04 **Multiple Logistic Regression**

Example 2 — Multiple Logistic Regression

Using Python Coding Method

Multiple Logistic Regression extends simple logistic regression by using **more than one independent variable** to predict a binary dependent variable.

Variables

Independent variables:

- Study_Hours
- Attendance_Percent
- Assignments_Completed
- Previous_Score
- Sleep_Hours

Dependent variable:

- Pass (0 = Fail, 1 = Pass)

Questions

Question 1 — Identifying the Variables

1. What is the dependent variable?
 2. What are the independent variables?
 3. Why is `Pass` suitable as the dependent variable for logistic regression?
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Question 2 — Formulating the Model

1. Write the multiple logistic regression model using the five independent variables.
 2. Express the model in terms of the **log-odds** of passing.
 3. Express the model in terms of the **probability of passing**.
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Question 3 — Estimating the Model

Using Python:

1. Import the required dataset.
 2. Define the dependent and independent variables.
 3. Estimate the multiple logistic regression model.
 4. Obtain the intercept and the five estimated coefficients.
 5. Display the estimated coefficients in a table.
 6. Identify the sign of each coefficient.
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Question 4 — Interpreting the Coefficients

For each independent variable:

1. Interpret its estimated coefficient.
 2. Determine whether an increase in the variable increases or decreases the probability of passing.
 3. Calculate the corresponding **odds ratio**.
 4. Interpret the odds ratio.
 5. Identify which variables are statistically significant.
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Question 5 — Writing the Estimated Model

Using the coefficients obtained from Python:

1. Write the estimated logistic regression equation in **log-odds form**.
 2. Write the corresponding probability equation.
 3. Explain the role of the intercept in the estimated model.
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Question 6 — Predicting the Probability of Passing

Consider the following student:

Variable	Value
Study_Hours	6

Attendance_Percent	85
Assignments_Completed	8
Previous_Score	70
Sleep_Hours	7

1. Calculate the value of the linear predictor (Z).
 2. Calculate the student's predicted probability of passing.
 3. Express the probability as a percentage.
 4. Interpret the result.
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Question 7 — Classification

Using a classification threshold of **0.5**:

1. Classify the student as **Pass** or **Fail**.
 2. Explain why the threshold of 0.5 is used.
 3. What would happen to the classification if the predicted probability were 0.45?
 4. What would happen if it were 0.75?
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Question 8 — Confusion Matrix

Using the testing dataset:

1. Generate the predicted classes.
 2. Construct the confusion matrix.
 3. Identify:
 - True Positives (TP)
 - True Negatives (TN)
 - False Positives (FP)
 - False Negatives (FN)
 4. Explain what each element means in the context of predicting student success.
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Question 9 — Classification Performance

Using the testing results, calculate:

1. **Accuracy**
2. **Precision**
3. **Recall**
4. **F1-score**

Then:

5. Interpret each measure in the context of predicting whether students pass.
 6. Which measure would be particularly important if the university wants to identify students who are actually at risk of failing?
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Question 10 — Comparison with Simple Logistic Regression

Compare the **Simple Logistic Regression** model with the **Multiple Logistic Regression** model.

1. What is the main difference between the two models?
 2. How many independent variables does each model use?
 3. How does the interpretation of a coefficient differ between the two approaches?
 4. Why might multiple logistic regression provide more information about student performance?
 5. Can a variable appear important in simple logistic regression but become less important after other variables are included?
 6. Which model would be more appropriate when several factors simultaneously influence the probability of passing?
 7. Compare the predictive performance of the two models using:
 - Accuracy
 - Precision
 - Recall
 - F1-score
 - Log Loss
 8. Based on the results, determine which model provides better predictive performance and justify your conclusion.
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